Lab 7: Fuzz Testing for Input Validation Bugs

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Summary of Identified Bugs and Validation Fixes

Property-based fuzz testing with Hypothesis and pytest exposed missing input validation across all three functions in the starter *Lab 7/train_fare.py* module. Invalid inputs triggered unhandled *TypeError*, *AttributeError*, *IndexError*, and *ValueError* exceptions (including Python's 4300-digit integer conversion limit), or returned negative/unsorted distances and malformed berth tuples.

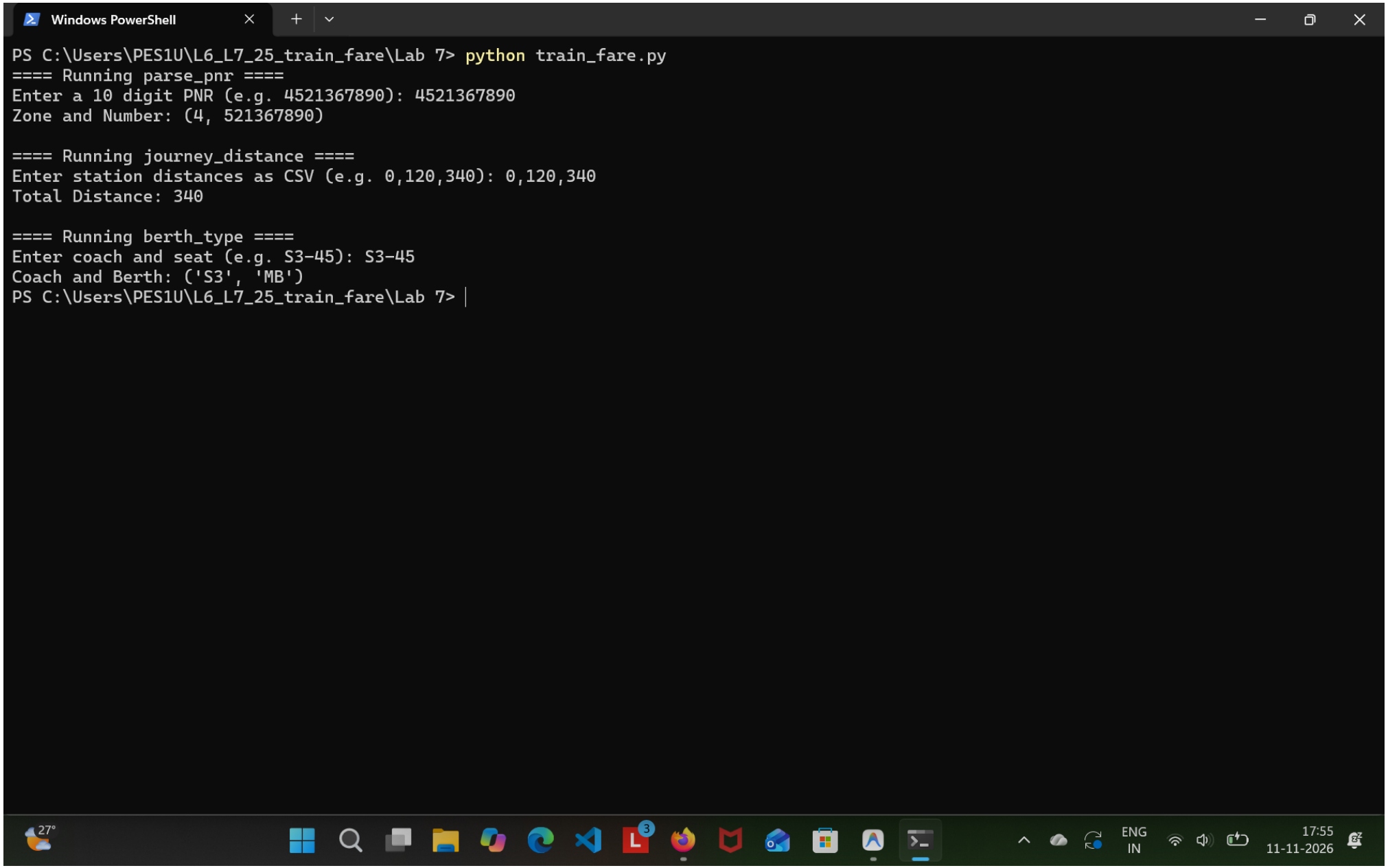
Function	Original Issues Exposed by Fuzzing	Implemented Validation in Fixed Code
parse_pnr(pnr)	Crashed with TypeError/IndexError/ValueError on non-strings, empty or wrong-length strings, outer spaces, signs (+/-), internal spaces, Unicode digits, and >=4300-digit strings.	Checks *isinstance(pnr, str)*, strips outer whitespace, enforces exact length 10 and strict ASCII digits ('0'-'9') via *_is_ascii_digits* before *int()*; returns None if invalid.
journey_distance(distances_csv)	Crashed on non-strings, empty fields ('0,,340'), Unicode numerals, signs, internal spaces, and >=4300-digit numbers; returned negative values on descending station inputs ('300,100').	Checks *str* type, non-empty input, <=1000 comma-separated parts, <=10 ASCII digits per token before *int()*, and non-decreasing station sequence (*value* >= *prev*); returns None if invalid.
berth_type(seat)	Crashed when '-' was missing or repeated, on non-strings, Unicode digits, or >=4300-digit seat numbers; accepted internal whitespace (' 0'), non-alphanumeric coaches, and seat numbers <= 0.	Checks *str* type, exact 2-part '-' split with unstripped internal spacing, 1-10 char ASCII alphanumeric coach code, 1-6 ASCII digits before *int()*, and seat_number > 0; returns None if invalid.

Scope and Verification Summary

• **Initial Fuzz Run:** 17 failed, 1 passed out of 18 Hypothesis property tests on the starter code.
• **Final Fuzz Run:** 18 passed in 1.05s out of 18 tests across all three functions after fixes.
• **Scope Note:** Validation checks input types, formatting, character set (strict ASCII digits/alphanumerics), length limits, positive seat integers, and monotonic station distances, returning *None* on invalid inputs. It does not verify domain-specific PNR checksums, actual railway station tables, or real coach seat capacities.

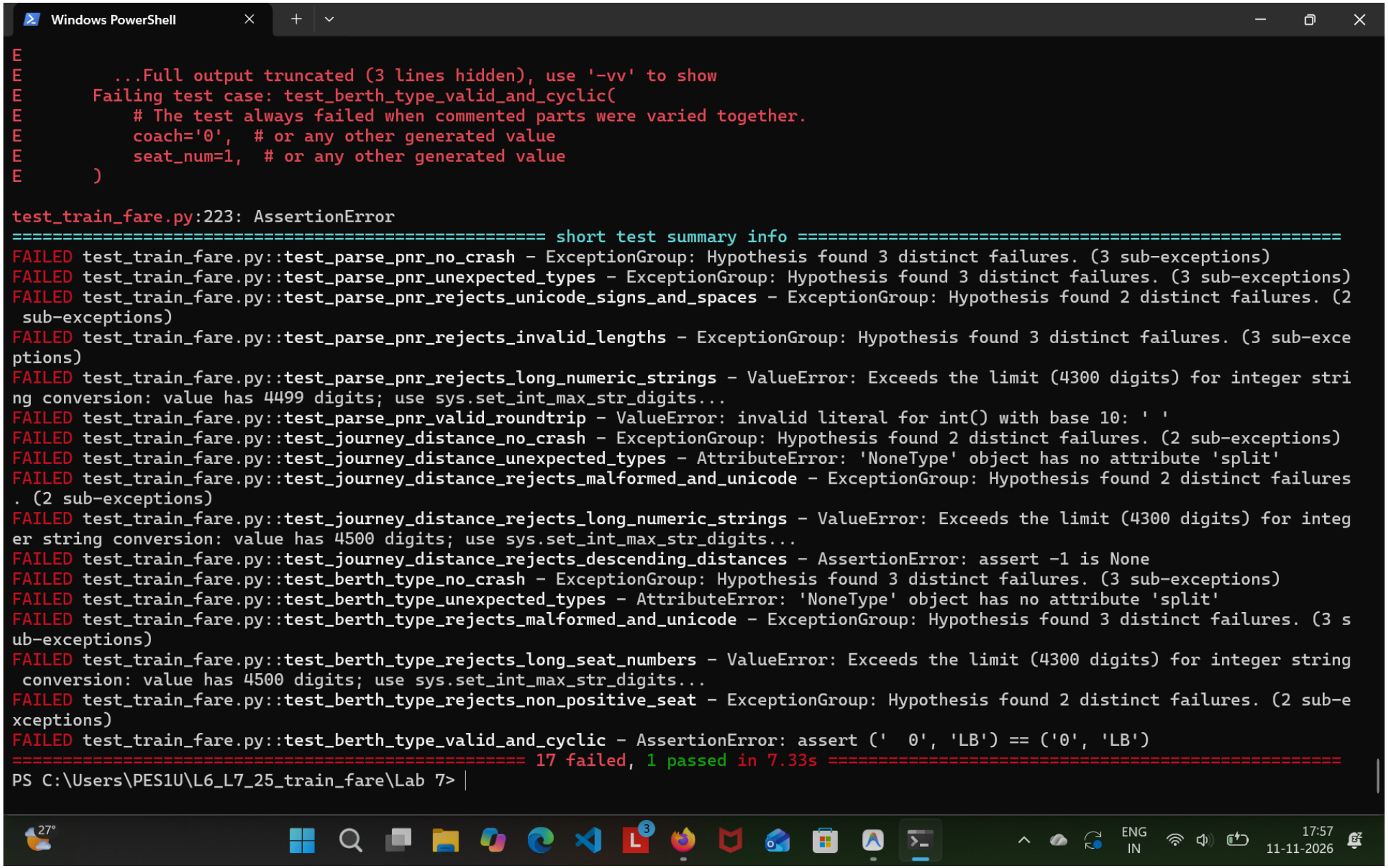
Stage 1: Original Application Execution (python train_fare.py)

Executed the unmodified starter script with nominal sample inputs (4521367890, 0,120,340, and S3-45). Normal inputs succeed as intended while hiding lack of defensive validation.



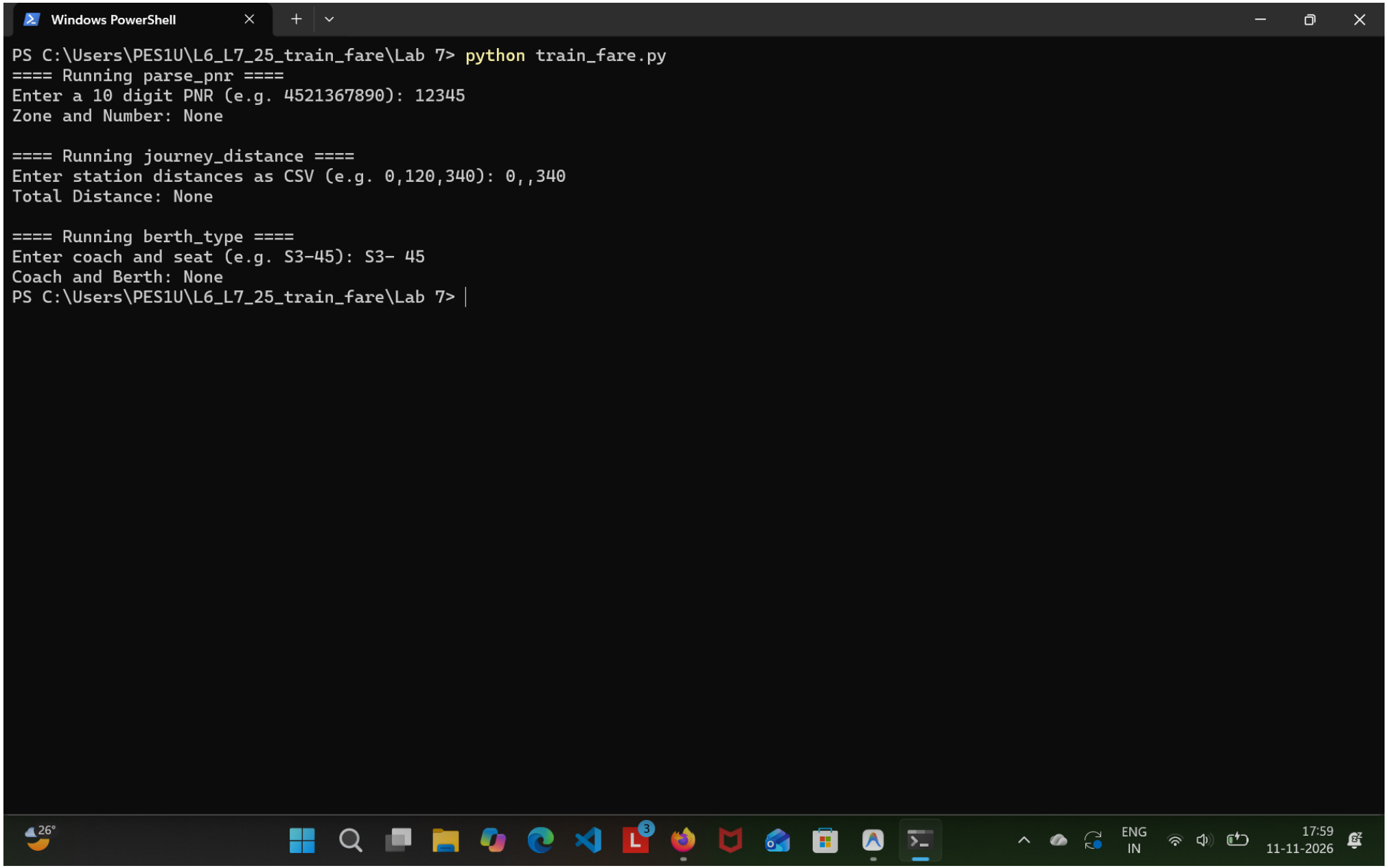
Stage 2: Initial Hypothesis Fuzz Tests Failing (python -m pytest test_train_fare.py -v)

Executed the 18-test Hypothesis suite against the unmodified starter code. Testing exposed 17 failures across parse_pnr, journey_distance, and berth_type (17 failed, 1 passed).



Stage 3: Fixed Application Execution (python train_fare.py)

Executed the hardened train_fare.py interactively with invalid and malformed inputs (12345, 0,,340, and S3- 45). Each function rejects invalid input safely by returning None.



Stage 4: Final Passing Hypothesis Suite (python -m pytest test_train_fare.py -v)

Re-ran the complete 18-test Hypothesis property-based suite against the fixed implementation. All 18 tests passed in 1.05s with zero failures or skipped cases.

